

Optimizing UAV-UGV System for Reconnaissance and Strike Based on Soft Actor Critic Algorithm

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Abstract: In this paper, reinforcement learning is used to solve the reconnaissance and strike tasks of UGVs and UAVs in the battlefield environment, and different types of agents are trained separately under the premise of environmental interaction. As a mature reinforcement learning algorithm, SAC algorithm has been widely used in the training of single agents. By assigning part of the state of the UGV to the UAV, the problem that the fully distributed structure does not converge is solved. For the actual battlefield environment, UGVs and UAVs need to undertake different tasks. The UAV provides accompanying vigilance for UGV and the main reconnaissance task of the target area. The UGV should go to different target areas according to task priority order to carry out auxiliary reconnaissance and strike tasks after reconnaissance.

Key Words: Reinforcement learning, Soft Actor-Critic, UGV and UAV system

1 Introduction

In recent years, the rapid advancement of unmanned systems has significantly reshaped modern warfare and complex mission planning. Among these systems, Unmanned Aerial Vehicles (UAVs) and Unmanned Ground Vehicles (UGVs) have garnered considerable attention due to their complementary capabilities. UAVs, with their superior mobility and wide-area surveillance capabilities, excel in reconnaissance, target detection, and real-time situational awareness. UGVs, on the other hand, possess advantages in persistent ground operations, precision engagement, and logistics, making them indispensable in a variety of mission scenarios. By combining these strengths, UAV-UGV collaborative systems offer a powerful solution for achieving efficient, coordinated, and intelligent strike operations.

The integration of UAVs and UGVs for cooperative missions poses several critical challenges ([1]-[3]). These include the design of robust coordination strategies ([4]), seamless communication protocols, and adaptive decision-making frameworks to address dynamic and uncertain environments. Furthermore, ensuring efficient resource allocation, minimizing operational latency, and maximizing mission success in high-stakes scenarios are key areas of focus. In the context of collaborative strike missions, the ability of UAVs to scout and identify targets in real-time, coupled with the precision of UGVs to engage these targets effectively, exemplifies the potential of such systems to outperform traditional unmanned operations ([5]). Recent advancements in artificial intelligence (AI), particularly in reinforcement learning, have provided promising avenues for addressing these challenges.

Reinforcement Learning (RL) has emerged as a powerful paradigm for enabling intelligent agents to learn optimal behaviors through interaction with their environments. By maximizing cumulative rewards, RL algorithms allow agents to autonomously discover strategies that balance exploration and exploitation.[6] Among the most notable advancements are model-free RL methods, such as Q-learning([7]) and

Policy Gradient approaches, which have demonstrated success in high-dimensional and continuous action spaces. More recently, algorithms like Deep Deterministic Policy Gradient (DDPG)([8]-[10]) and Soft Actor-Critic (SAC) ([11],[12]), have leveraged deep neural networks to extend RL to complex tasks involving continuous control, offering greater stability and sample efficiency.

This paper investigates a novel framework for UAV-UGV collaborative strike missions, focusing on optimizing their coordination through advanced reinforcement learning techniques. Specifically, we design a system where UAVs perform real-time reconnaissance to identify and classify targets, while UGVs execute precise strike actions informed by the UAV-provided intelligence. Experimental results show that the framework can select the maneuvering path and attack point automatically, and realize the good cooperation ability of UAV and UGV.

2 Method

In this section, we first present the reconnaissance strike problem of the UGV-UAV system based on the actual battlefield environment, and subsequently analyze the problem and jointly deal with it with the SAC-based method.

2.1 Problem Description

For the actual modern battlefield environment, consider a specific scenario: an unmanned vehicle and UAV system to complete the reconnaissance of the target in the enemy area on the map and select the appropriate strike point to fire artillery after the reconnaissance.

Given a sequence of target regions Γ , where each target area T_i has a center, priority, shape characteristics and target point. For each target area, the center address is defined as $c_i = (x_i, y_i)$, where x_i and y_i are the coordinates of the center of the target area in the 2D space, and the priority p_i is a scalar representing the priority of the target area. Higher values indicate higher priority. And for the shape parameters, we defined rectangle area whose length is l_i and width is w_i . In addition, each target region has a target point t_i corresponding to it. The parameters of each target point

include address $c_i = (x_i, y_i)$, priority p_i and threat degree d_i . So, the target area can be defined as $T_i = ((x_i, y_i), p_i, l_i, w_i, t_i)$, $t_i = ((x_i, y_i), p_i, d_i)$.

The state of the UGV is defined as

$$s_g = (x_g, y_g, v_{g_x}, v_{g_y}, \theta_g, \alpha_{g_s}, \beta_{g_s}, \alpha_{g_{s_{max}}}, \beta_{g_{s_{max}}}, d_{g_{scan}}, d_{g_{strike}}, s_{g_c}, s_{g_s}) \quad (1)$$

where (x_g, y_g) is the current position, (v_{g_x}, v_{g_y}) is horizontal and vertical velocity, $(\theta_g, \alpha_{g_s}, \beta_{g_s})$ are the orientation angle of the UGV, the reconnaissance angle and the artillery attack angle. $(\alpha_{g_{s_{max}}}, \beta_{g_{s_{max}}}, d_{g_{scan}}, d_{g_{strike}})$ are fixed parameters and are the maximum reconnaissance and strike angle and the maximum reconnaissance and strike distance of the UGV. (s_{g_c}, s_{g_s}) are UGV reconnaissance device and artillery status.

The state of the UAV is defined as

$$s_a = (x_a, y_a, z_a, v_{a_x}, v_{a_y}, v_{a_z}, \phi_a, \beta_a, \varphi_a, d_{a_{scan}}) \quad (2)$$

where (x_a, y_a, z_a) is the current position, $(v_{a_x}, v_{a_y}, v_{a_z})$ is velocities in x , y and z directions, $(\phi_a, \beta_a, \varphi_a)$ is the attitude angle of the UAV (roll angle, pitch angle, yaw angle). $d_{a_{scan}}$ is the maximum reconnaissance radius of the UAV.

2.2 UGV-UAV System

Considering the actual battlefield environment, we discuss UGV-UAV system within the continuous action framework. For any agent in the system, its action and state space are continuous. We define tuples (S, A, p, r) which is a Markov decision process (MDP), where S and A are action and state space. $p(s'|s, a)$ describes the probability density that the environment will transition from one state s to another s' after the agent takes a certain action a . $r(s, a)$ defines the agent in a certain state s to perform action a and the immediate reward received when a . This reward can be a real number indicating how good or bad the action was.

SAC introduces the concept of maximum entropy whose goal is to learn a policy that maximizes a weighted sum of reward and entropy. Its objective function is defined as

$$J(\pi) = \bar{\mathbf{a}}_{s_t \sim p^\pi, a_t \sim \pi} \left[r_t + \alpha \cdot \bar{\mathbf{a}}_{s_{t+1} \sim p} \left[\log \pi(a_t | s_t) - Q(s_t, a_t) \right] \right] \quad (3)$$

where r_t is the immediate reward received after the agent performs action a_t in state s_t . α is a hyperparameter that regulates the entropy weight and controls the balance between reward and entropy. Compared to the previous algorithm, entropy addition gives the agent a better exploration-exploitation balance. $Q(s_t, a_t)$ is a Q-value function of the current state s_t and action a_t and represents the expected total reward under this state-action pair. $\pi(a_t | s_t)$ is the policy function that represents the probability of choosing an action a_t in state s_t . p is the state transition probability of the environment.

The SAC algorithm has become an important tool in the field of reinforcement learning due to its characteristic of maximizing entropy, its ability to adapt to continuous and high-dimensional spaces, as well as its strong sample efficiency and stability. Especially when it comes to balancing exploration and exploitation, solving complex tasks and high-dimensional problems, SAC demonstrates superior performance. The method of SAC not only plays a role in single-agent reinforcement learning but also performs well in multi-agent reinforcement learning in the algorithm MASAC ([13]). And if the strategy of centralized training and Decentralized Execution (CTDE) is used, the information is centralized and the strategy of the agents is jointly updated during training, and the agents make decisions independently based on their local information and strategy during execution. CTDE is a commonly used framework in multi-agent systems, where at training time all agents share information about the environment and train jointly, but each agent makes decisions based on its own observations at execution time. In this way, despite their different state and action space dimensions, the overall system behavior can be optimized by designing an appropriate joint learning strategy. For the system of single UGV and single UAV, the joint state dimension of unmanned vehicle and UAV will bring large computational overhead.

If we consider updating the state in the environment at the same time, but instead of using centralized training, we generate their own experience pools, sample and train them separately, and then execute them separately. In the system of UGV and UAV, the high mobility of UAV can help UGV to complete the reconnaissance of the target area, and the enemy units in the target area can be detected earlier, so that the UGV can strike in a short time. At the same time, on the way, the UAV can learn the information of the UGV, which can provide a charging platform for the UAV.

The overall structure is shown in Fig. 1:

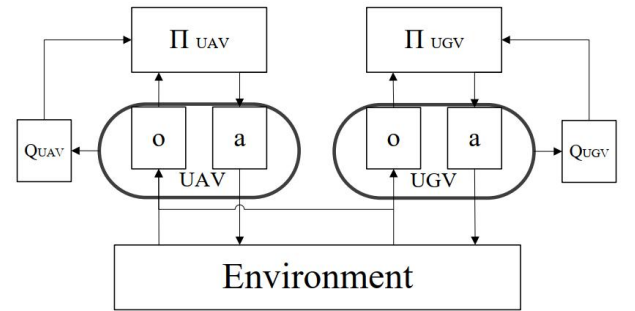


Fig. 1: The framework

Compared with the fully distributed structure that is difficult to converge, the UAV can obtain part of the information (position and reconnaissance strike status) of the UGV, so that it can overcome the impact of the dynamic nature of part of the environment in the face of the change of the UAV. The specific update process for the UGV-UAV system is as follows, the movement of the UGV is:

$$\begin{cases} v_{g_{x+1}} = v_{g_x} + \Delta t \cdot a_{g_x} \\ v_{g_{y+1}} = v_{g_y} + \Delta t \cdot a_{g_y} \\ x_{g_{x+1}} = x_{g_x} + \Delta t \cdot v_{g_x} \\ x_{g_{y+1}} = x_{g_y} + \Delta t \cdot v_{g_y} \end{cases} \quad (4)$$

the movement of the UGV is:

$$\begin{cases} v_{a_{x+1}} = v_{a_x} + \Delta t \cdot a_{a_x} \\ v_{a_{y+1}} = v_{a_y} + \Delta t \cdot a_{a_y} \\ v_{a_{z+1}} = v_{a_z} + \Delta t \cdot a_{a_z} \\ x_{a_{x+1}} = x_{a_x} + \Delta t \cdot v_{a_x} \\ x_{a_{y+1}} = x_{a_y} + \Delta t \cdot v_{a_y} \\ x_{a_{z+1}} = x_{a_z} + \Delta t \cdot v_{a_z} \end{cases} \quad (5)$$

When the UGV-UAV system carries out the reconnaissance phase of the target area, the target area is divided into several squares in order to determine the scanning progress.

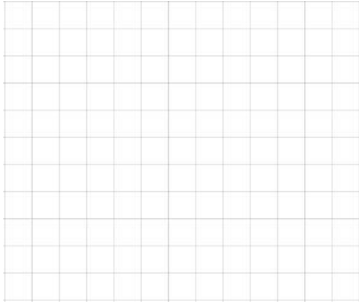


Fig. 2 Target Area Segmentation

The UGV and the UAV will adopt corresponding scanning methods respectively. The UAV will scan the circle with a certain radius $d_{a_{scan}}$, and the UGV will scan the front through its own device, The scanning range is a sector of radius $d_{g_{scan}}$.

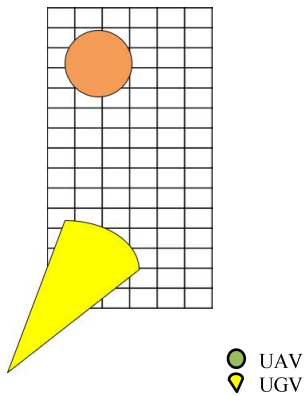


Fig. 3 Target Area Scanning

The target subregion will be marked as a scouted state after it has been scouted for a fixed time, enemy units will be

distributed in a sub-area, and when the UGV-UAV system has completely scanned the entire area, it can attack. For a specific target, the UGV will calculate its current strike success rate to decide when to strike, and the strike success rate is calculated as follows:

$$s_{rate} = d_{UGV} \cdot d_{t_i} \cdot \beta_{g_s} \quad (6)$$

where s_{rate} is the current strike success rate of the specific target, d_{UGV} , d_{t_i} , β_{g_s} are the normalized distance between the UGV and enemy units, the threat degree of enemy units, and the artillery direction.

The objective function in the update process is as follows,

$$J_{UGV}(\pi) = E_{s_t \sim p} \pi_{UGV, a_t \sim \pi_{UGV}} [r_{UGV} + \alpha_{UGV} \cdot E_{s_{t+1} \sim p} [\log \pi_{UGV} - Q(s_t, a_t)]] \quad (7)$$

$$J_{UAV}(\pi) = E_{s_t \sim p} \pi_{UAV, a_t \sim \pi_{UAV}} [r_{UAV} + \alpha_{UAV} \cdot E_{s_{t+1} \sim p} [\log \pi_{UAV} - Q(s_t, a_t)]] \quad (8)$$

The respective Q-network update formulas for both are as follows,

$$\begin{aligned} Q_{1,UGV}(s_t, a_t) = & r_t + \gamma E_{s_{t+1}} [\min(Q'_{1,UGV}(s_{t+1}, a_{t+1}), Q'_{2,UGV}(s_{t+1}, a_{t+1})) \\ & - \alpha_{UGV} \log \pi_{UGV}(a_{t+1} | s_{t+1})] \end{aligned} \quad (9)$$

$$\begin{aligned} Q_{2,UGV}(s_t, a_t) = & r_t + \gamma E_{s_{t+1}} [\min(Q'_{1,UGV}(s_{t+1}, a_{t+1}), Q'_{2,UGV}(s_{t+1}, a_{t+1})) \\ & - \alpha_{UGV} \log \pi_{UGV}(a_{t+1} | s_{t+1})] \end{aligned} \quad (10)$$

$$\begin{aligned} Q_{1,UAV}(s_t, a_t) = & r_t + \gamma E_{s_{t+1}} [\min(Q'_{1,UAV}(s_{t+1}, a_{t+1}), Q'_{2,UAV}(s_{t+1}, a_{t+1})) \\ & - \alpha_{UAV} \log \pi_{UAV}(a_{t+1} | s_{t+1})] \end{aligned} \quad (11)$$

$$\begin{aligned} Q_{2,UAV}(s_t, a_t) = & r_t + \gamma E_{s_{t+1}} [\min(Q'_{1,UAV}(s_{t+1}, a_{t+1}), Q'_{2,UAV}(s_{t+1}, a_{t+1})) \\ & - \alpha_{UAV} \log \pi_{UAV}(a_{t+1} | s_{t+1})] \end{aligned} \quad (12)$$

The reward function of the UAV is designed as

$$R_{UAV} = \lambda_1 \cdot R_{approach} + \lambda_2 \cdot R_{scan} \cdot k_{order} + 0.8 \cdot \lambda_3 \cdot R_{strike} \cdot k_{order} + \lambda_4 \cdot R_{done} + \lambda_5 \cdot R_{collision} \quad (13)$$

Where λ_1 , λ_2 , λ_3 , λ_4 and λ_5 are the coefficients of various reward values. $R_{approach}$ is the reward for being close to the targets. R_{scan} is the reward for scanning the targets. k_{order} is the order coefficient of reconnaissance or strike, in this way, the reward differentiation under different priorities can be achieved. R_{strike} is the reward for striking the targets by the UGV. R_{done} is a one-time reward for completing the task. $R_{collision}$ is the collision penalty of the UAV, including the penalty of touching the boundary.

The reward function of the UGV is designed as

$$R_{UGV} = \lambda_1 \cdot R_{approach} + \lambda_2 \cdot R_{scan} \cdot k_{order} + \lambda_3 \cdot R_{strike} \cdot k_{order} + \lambda_4 \cdot R_{done} + \lambda_5 \cdot R_{collision} \quad (14)$$

Where λ_1 , λ_2 , λ_3 , λ_4 and λ_5 are the coefficients of various reward values. $R_{approach}$ is the reward for being close to the targets. R_{scan} is the reward for scanning the targets. k_{order} is the order coefficient of reconnaissance or strike, in this way, the reward differentiation under different priorities can be achieved. R_{strike} is the reward for striking the targets

by the UGV. R_{done} is a one-time reward for completing the task. $R_{collision}$ is the collision penalty of the UGV, including the penalty of touching the boundary.

3 Experiments

The experiment designs two target areas, which are randomly assigned to an enemy unit, and the UGV and UAV set off from the starting point to complete the reconnaissance and strike tasks.

The reward changes during training are shown below,

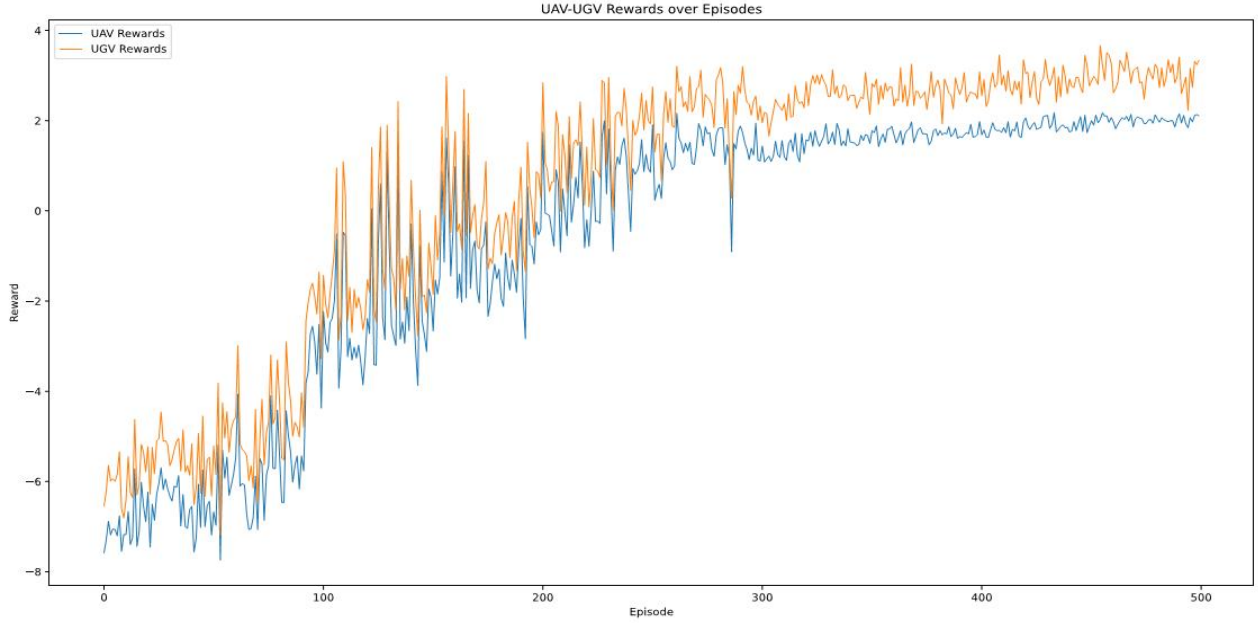


Fig. 4 Reward Function Curve

The task of training and applying the trained network test is detailed below,

priority of the target area, and go to complete the reconnaissance mission together.

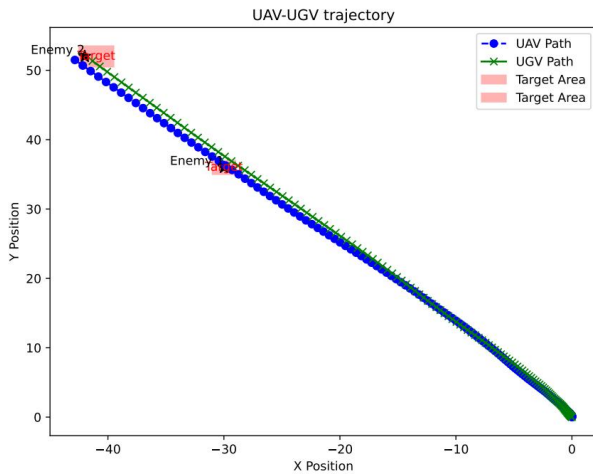


Fig. 5 UAV-UGV Trajectory-1

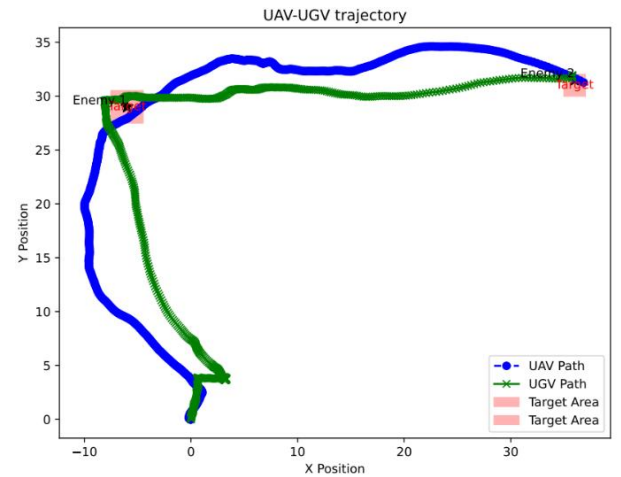


Fig. 6 UAV-UGV Trajectory-2

Figs. 5 and 6 depict the trajectories of the unmanned vehicle-UAV system in the reconnaissance and strike missions under different maps. The UGV and the UAV together start from the (0,0) point at the same time, then the UGV and UAV make order judgment according to the

Figure 5 designs two target regions with simple arrangement, the center points are (-30,35) and (-43,52), respectively, and the combinations of length and width are (2,3) and (3,6), respectively. The enemy units are randomly generated in the target areas.

Figure 6 designs two target regions with simple arrangement, the center points are (36,31) and (-6,29), respectively, and the combinations of length and width are (2,2) and (3,3), respectively. The enemy units are randomly generated in the target areas in the same way.

By incorporating part of the state of the UGV into the state space of the UAV for training, the experiment result shows that the UGV-UAV system can successfully complete the task.

4 Conclusion

This paper studies the UGV-UAV system in the battlefield environment to complete the reconnaissance and attack task of enemy units in multiple target areas. The UGV and UAV need to judge the attack sequence and conduct reconnaissance and strike work in turn. The UAV undertakes the main reconnaissance task in the target area, and the unmanned vehicle undertakes the auxiliary reconnaissance and strike tasks. By assigning part of the state of the unmanned vehicle to the UAV, the problem of non-convergence of the fully distributed structure is solved. In the current study, the environment design for the actual battlefield is relatively simple and does not take into account the complex terrain and environment of the battlefield, which can be improved accordingly later. At the same time, the evaluation dimension of task completion is currently limited to whether the task is completed, mission completion time, accuracy of target identification, and resource utilization efficiency should be added in the follow-up study.

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